

0.1 Abel's Formular and Abel's Test

0.1.1 Abel's Formular: Summation by Parts

Lemma 0.1.1. Given sequences $\{a_n\}$ and $\{b_n\}$, denote $A_n = \sum_{k=0}^n a_k$.
Then, for any $q \in \mathbb{N}$,

$$\sum_{n=0}^q a_n b_n = \sum_{n=0}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q = \sum_{n=0}^{q-1} \left(\sum_{k=0}^n a_k \right) \cdot (b_n - b_{n+1}) + b_q \cdot \sum_{n=0}^q a_n.$$

In particular, if $0 \leq p \leq q$,

$$\sum_{n=p}^q a_n b_n = \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p.$$

Proof. Since $a_n = A_n - A_{n-1}$ (with $A_{-1} \stackrel{\text{def}}{=} 0$),

$$\begin{aligned} \sum_{n=0}^q a_n b_n &= \sum_{n=0}^q (A_n - A_{n-1}) b_n = \sum_{n=0}^q A_n b_n - \sum_{n=0}^q A_{n-1} b_n \\ &= \sum_{n=0}^q A_n (b_n - b_{n+1}) + \sum_{n=0}^q A_n b_{n+1} - \sum_{n=0}^q A_{n-1} b_n \\ &= \sum_{n=0}^q A_n (b_n - b_{n+1}) + A_q b_{q+1} \\ &= \sum_{n=0}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q. \end{aligned}$$

Particular,

$$\begin{aligned} \sum_{n=p}^q a_n b_n &= \sum_{n=0}^q a_n b_n - \sum_{n=0}^{p-1} a_n b_n \\ &= \sum_{n=0}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - \left(\sum_{n=0}^{p-2} A_n (b_n - b_{n+1}) + A_{p-1} b_{p-1} \right) \\ &= \sum_{n=p-1}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_{p-1} \\ &\quad \hline &= \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_{p-1} b_{p-1} - A_{p-1} b_p + A_q b_q - A_{p-1} b_{p-1} \\ &= \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p. \end{aligned}$$

□

Note: Denote $A_n = \sum_{k=0}^n a_k$ and $\Delta b_n = b_{n+1} - b_n$, then

$$\sum_{n=0}^q a_n b_n = A_q b_q - \sum_{n=0}^{q-1} A_n \Delta b_n$$

It is similar to:

$$\int f \cdot g = Fg - \int Fg'$$

0.1.2 Abel's Test

Theorem 0.1.2. If $\sum a_n$ converges and $\{b_n\}$ is bounded and monotone sequence, then $\sum a_n b_n$ converges.

Proof. Since $\{b_n\}$ is bounded and monotone, it is convergent. Put $b = \lim_{n \rightarrow \infty} b_n$ and $A_n = \sum_{k=0}^n a_k$, $A_{-1} = 0$. WLOG, suppose $\{b_n\}$ is monotone decreasing. Then $\{b_n - b\}$ converges to zero and is decreasing.

Let $M > 0$ be a bound of $|A_n|$. Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ such that $n \geq N \implies |b_n - b| < \frac{\varepsilon}{2M}$.

By Abel's formula, for all $q \geq p \geq N$,

$$\begin{aligned} \left| \sum_{n=p}^q a_n(b_n - b) \right| &= \left| \sum_{n=p}^{q-1} A_n(b_n - b_{n+1}) + A_q(b_q - b) - A_{p-1}(b_p - b) \right| \\ &\leq M \cdot \left[\sum_{n=p}^{q-1} \underbrace{|b_n - b_{n+1}|}_{\geq 0} + \underbrace{|b_q - b|}_{\geq 0} + \underbrace{|b_p - b|}_{\geq 0} \right] \\ &= M \cdot \left[\sum_{n=p}^{q-1} (b_n - b_{n+1}) + (b_q - b) + (b_p - b) \right] \\ &= M \cdot [(b_p - b_q) + (b_q - b) + (b_p - b)] \\ &= 2M(b_p - b) < 2M \cdot \frac{\varepsilon}{2M} = \varepsilon \end{aligned}$$

Therefore, by the Cauchy Criterion for series, $\sum a_n(b_n - b)$ converges.

Moreover, since $\sum a_n$ converges, so is $b \cdot \sum a_n$. Finally,

$$\sum_{n=0}^{\infty} a_n(b_n - b) + \sum_{n=0}^{\infty} a_n \cdot b = \sum_{n=0}^{\infty} a_n b_n$$

converges as a sum of two convergent series. □

Note: If $b = 0$, it is sufficient for the convergence of $\sum a_n b_n$ that $\sum a_n$ be bounded.

In this case, $\sum a_n$ does not necessarily have to converge (it is called Dirichlet's Test).

Example. A Series

$$\sum_{n=1}^{\infty} \frac{\sin n}{n}$$

converges. Enough to show that: $\sum_{n=1}^N \sin n$ is bounded. Then we can apply Dirichlet's Test. Observe that

$$\begin{aligned} 2 \sin(1) \cdot \sum_{n=1}^N \sin n &= \sum_{n=1}^N 2 \sin(1) \sin n = \sum_{n=1}^N [\cos(n-1) - \cos(n+1)] \\ &= \cos 0 - \cos 2 + \cos 1 - \cos 3 + \cos 2 - \cos 4 + \cdots + \cos(N-2) - \cos(N) + \cos(N-1) + \cos(N+1) \\ &= \cos 0 + \cos 1 - \cos(N) - \cos(N+1) \end{aligned}$$

Therefore, for any $N \in \mathbb{N}$,

$$\left| \sum_{n=1}^N \sin n \right| = \left| \frac{\cos 0 + \cos 1 - \cos(N) - \cos(N+1)}{2 \sin(1)} \right| \leq \frac{2}{\sin(1)}$$

Now, Applying the Dirichlet's Test, we obtain the result. But, it is not converges absolutely.

Dirichlet-De la Vallée Poussin.

Let $f_n, g_n: X \rightarrow \mathbb{C}$ ($n \in \mathbb{N}$) be sequence of functions.

Suppose that

a). $F_n(x) \stackrel{\text{def}}{=} \sum_{k=1}^n f_k(x)$ is uniformly bounded.

b). $\sum |g_n - g_{n-1}|$ converges uniformly.

c). $g_n \rightarrow 0$ uniformly

Then, $\sum f_n g_n$ converges uniformly.

Proof. By the Abel's formula, given $x \in X$, and $0 < p < q$,

$$\begin{aligned} \sum_{n=p+1}^q f_n(x) g_n(x) &= \sum_{n=1}^q f_n(x) g_n(x) - \sum_{n=1}^p f_n(x) g_n(x) \\ &= F_q(x) \cdot g_q(x) - \sum_{n=1}^{q-1} F_n(x) \cdot (g_{n+1}(x) - g_n(x)) \\ &\quad - \left[F_p(x) g_p(x) - \sum_{n=1}^{p-1} F_n(x) \cdot (g_{n+1}(x) - g_n(x)) \right] \\ &= F_q(x) \cdot g_q(x) - F_p(x) \cdot g_p(x) - \sum_{n=p}^{q-1} F_n(x) \cdot [g_{n+1}(x) - g_n(x)] \end{aligned}$$

Now, for bound $M > 0$ of $\{F_n\}$,

$$\begin{aligned} \left| \sum_{n=p+1}^q f_n(x) g_n(x) \right| &= \left| F_q(x) g_q(x) - F_p(x) g_p(x) - \sum_{n=p}^{q-1} F_n(x) [g_{n+1}(x) - g_n(x)] \right| \\ &\leq M \cdot \left[|g_q(x)| + |g_p(x)| + \underbrace{\sum_{n=p}^{q-1} |g_{n+1}(x) - g_n(x)|}_{\text{Cauchy, by b.}} \right] \rightarrow 0. \end{aligned}$$